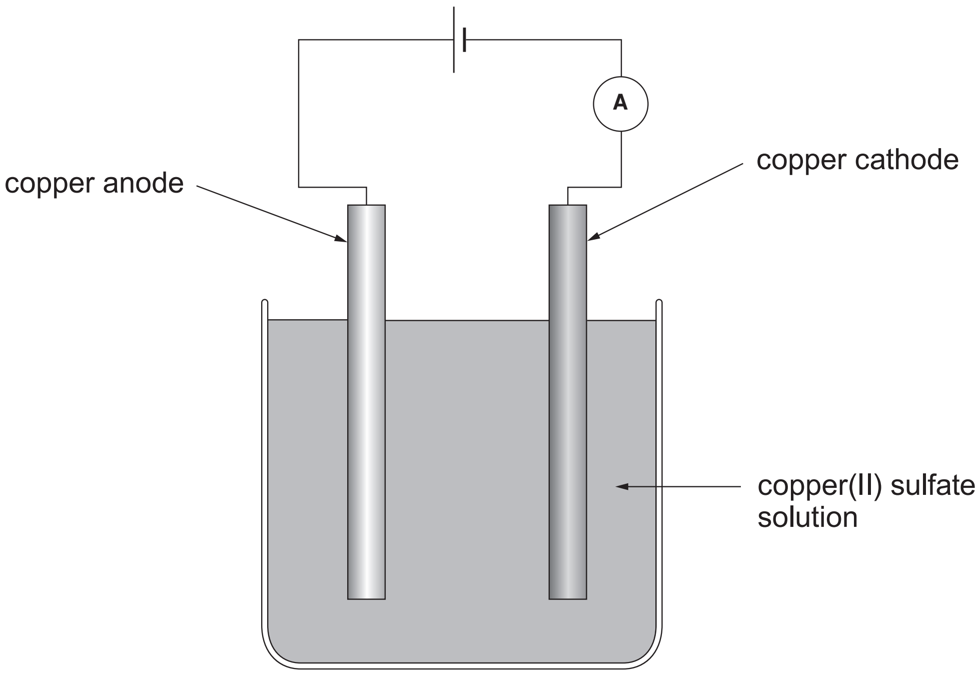
Examiner only

8. A student used the apparatus shown to investigate how changing current affects the mass of copper deposited on the cathode.



Equal volumes and concentrations of copper(II) sulfate solution were used each time. Each experiment was run for 10 minutes. All the readings were obtained at room temperature. The following procedure was followed before the mass of copper could be found.

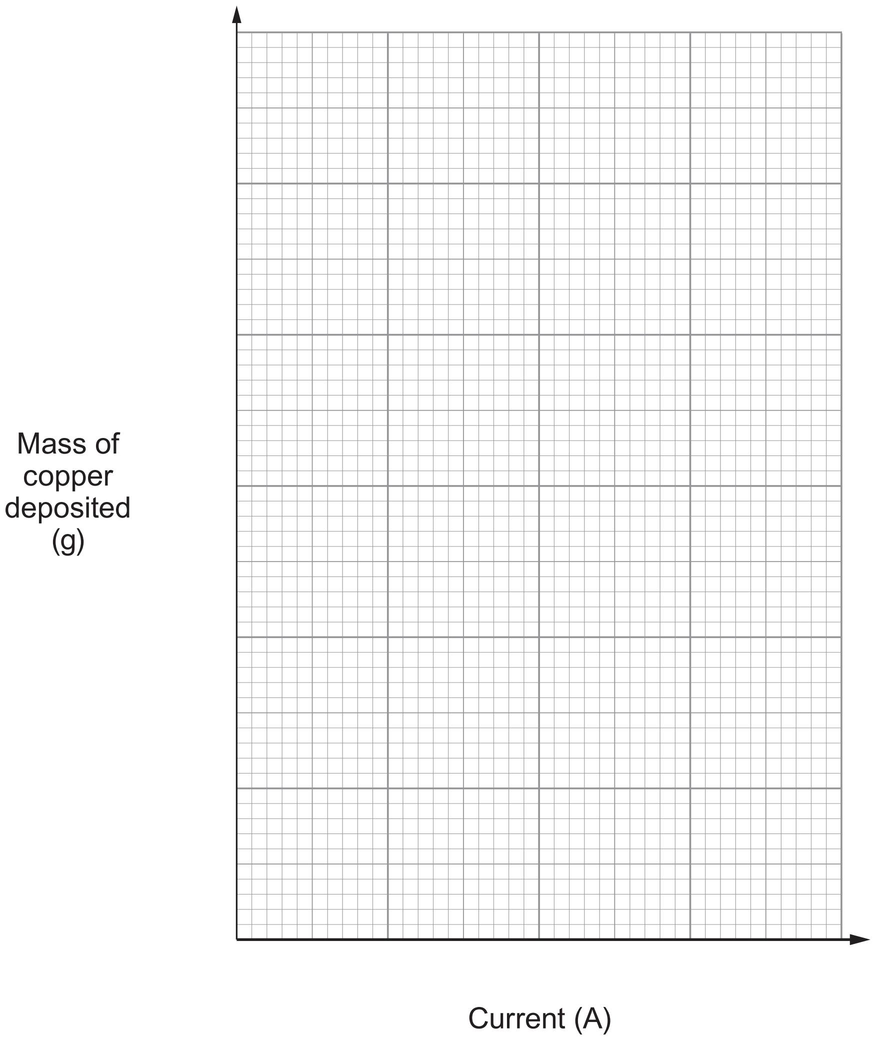
- the electrode was carefully removed from the electrolyte
- the copper deposit was washed
- the electrode and copper deposit were dried

Her results are shown in the following table.

Current (A)	Mass of copper deposited (g)
0.0	0.00
0.5	0.16
1.0	0.30
1.5	0.43
2.0	0.60

Examiner only

(a) Choose suitable scales for the axes below and plot the current against the mass of copper deposited. Draw a suitable line. [4]



(b) Use your graph to predict the mass of copper deposited using a current of 3.5 A. [1]

Mass of copper deposited = ... g

Examiner only

(c) Explain why the electrolyte keeps its blue colour during the electrolysis process. Include electrode equations to support your answer. [4]

....

....

....

....

....

....

9

Examiner only

8. Methanol, ethanol and butanol all belong to the homologous series of alcohols.

(a) The table shows information about two different methods for the manufacture of ethanol.

	Method **A** fermentation	Method **B** addition reaction
Raw material	sugar from sugar cane	ethene from crude oil
Reaction	yeast catalyst $C_6H_{12}O_6(aq) \longrightarrow 2C_2H_5OH(aq) + 2CO_2(g)$	phosphoric(V) acid catalyst $C_2H_4(g) + H_2O(g) \longrightarrow C_2H_5OH(l)$
Operating pressure	1 atm	60 atm
Type of process	batch (stop-start)	continuous (runs all the time)

Use the information in the table and your knowledge to answer the following question.

Explain **two** advantages and **two** disadvantages of method **A** compared with method **B**. [4]

Advantages

1. ....

....

2. ....

....

Disadvantages

1. ....

....

2. ....

....

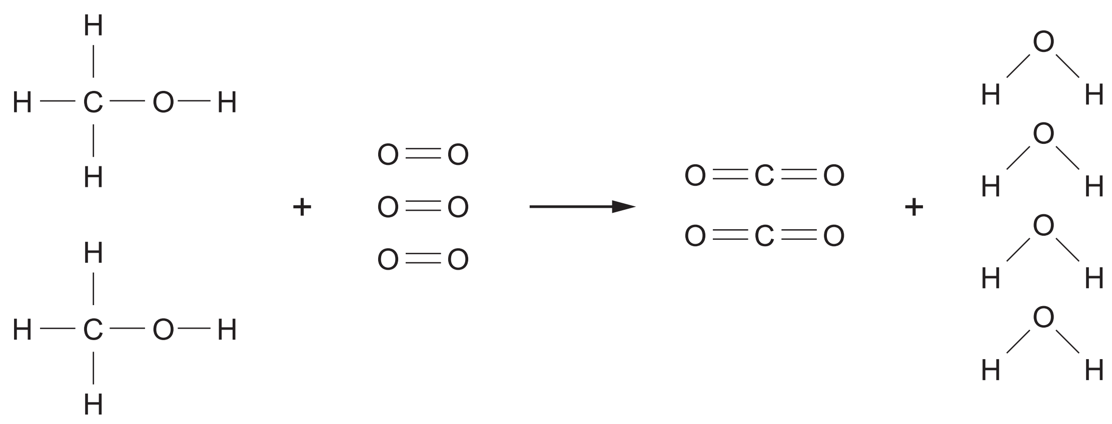
Examiner only

(b) Ethanol is found in alcoholic drinks. Alcoholic drinks turn sour when left exposed to air because ethanol is oxidised to ethanoic acid and water.

Complete the balanced symbol equation for this reaction. [1]

$C_2H_5OH \quad + \quad O_2 \longrightarrow$ ....

(c) Methanol readily burns in air. The diagram shows the bonds which are broken and the bonds which are formed during the combustion of methanol.



Some relevant bond energies are shown in the table.

Bond	Bond energy (kJ)
C—H	413
C—O	358
C═O	805
O—H	464

Examiner only

(i) The total energy needed to break all the bonds in the reactants is 5616 kJ. The energy needed to break the bonds in one molecule of methanol is 2061 kJ.

Use this information to calculate the amount of energy needed to break **one** O═O bond. [2]

Energy needed = ... kJ

(ii) Calculate the total energy released when all the bonds in the products are formed. [2]

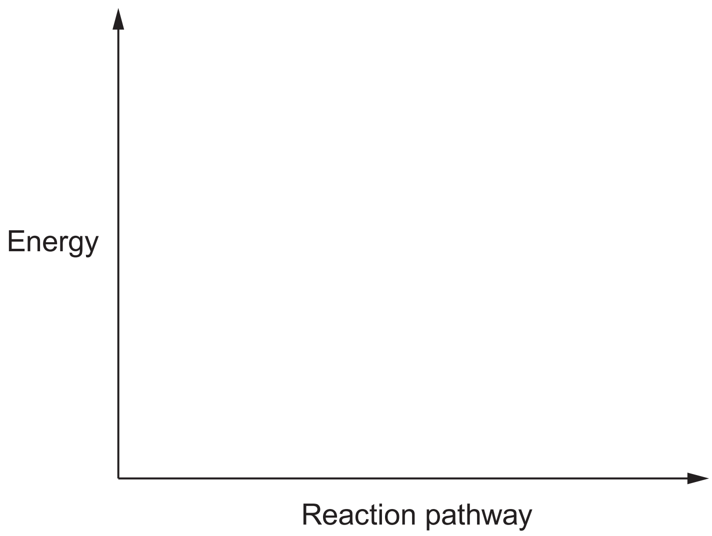
Energy released = ... kJ

(iii) The burning of methanol gives out heat and is said to be exothermic. Use the total energy value 5616 kJ and your answer to part (ii) to show that this is correct. [1]

....

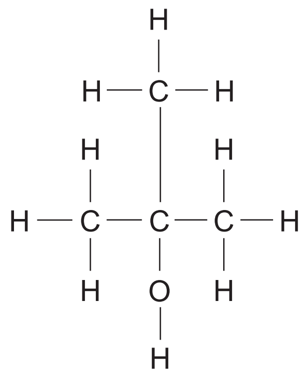
....

(iv) On the axes below draw the energy profile for the combustion of methanol and use the symbol ($\updownarrow$) to show the activation energy for the reaction. [1]

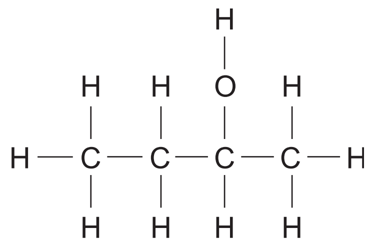


Examiner only

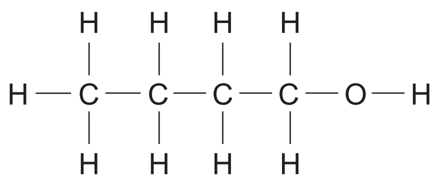
(d) Butanol has the molecular formula C_4H_9OH. It has four positional isomers, **A**, **B**, **C** and **D**.



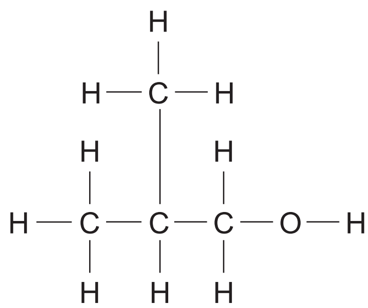
A



B



C



D

Give the **letter** of the isomer corresponding to each of the names in the table. [2]

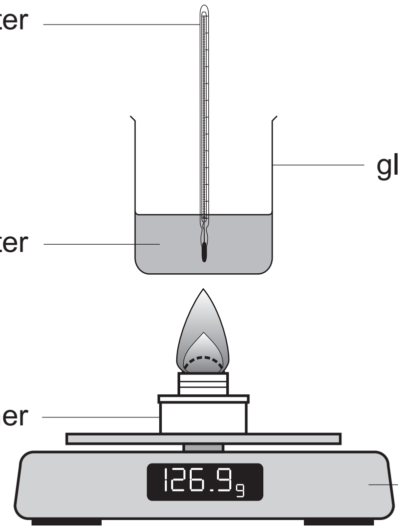
Name	Isomer
butan-1-ol	
butan-2-ol	
2-methylpropan-1-ol	
2-methylpropan-2-ol	

13

8. Mountaineers often choose an ethanol burner when hiking in extremely cold conditions.

ethanol burner

A group of students was asked to investigate how the mass of ethanol burned is related to the amount of energy given out.



Four students each burned a different mass of ethanol and recorded the temperature rise of 250 g of water. The diagram shows the apparatus used by each of the students.
They used their results to calculate the energy given out.

The table shows the students' results.

Student	Mass of ethanol burned (g)	Energy given out (J × 10^4)
A	1.1	2.2
B	1.8	3.6
C	2.9	5.8
D	4.2	8.4

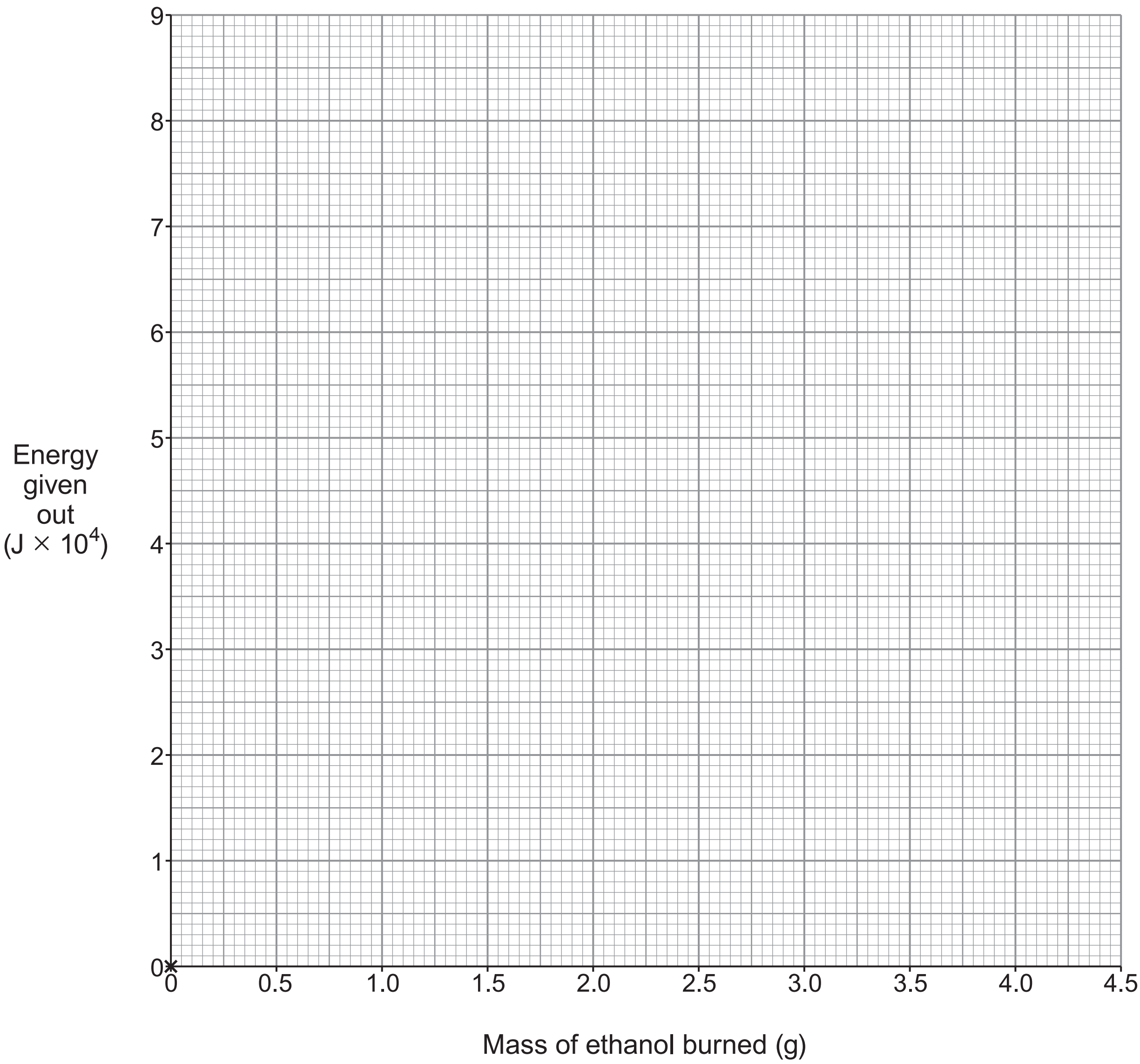
(a) The temperature rise can be calculated using the formula below.

energy given out (J) = 4.2 × temperature rise (°C) × mass of water (g)

Use this formula to calculate the temperature rise recorded by student **D**. [2]

Temperature rise = ... °C

(b) Plot the energy given out against mass of ethanol burned on the grid below and draw a suitable line. [3]



Examiner only

(c) Describe the relationship between the mass of ethanol burned and the energy given out. [2]

....

....

....

(d) All the recorded temperature rises were **lower** than expected.

Explain **one** piece of advice you would give the students to resolve this problem. [2]

....

....

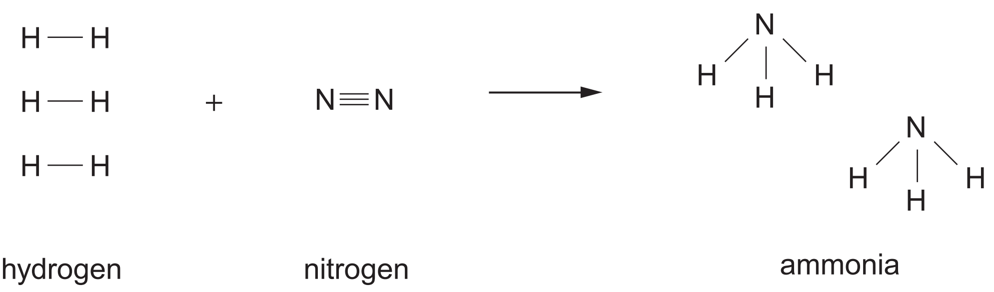
....

9

Examiner only

8. Ammonia is manufactured from hydrogen and nitrogen in the Haber process.

(a) The equation shows the bonds which are broken and the bonds which are formed in the production of ammonia.

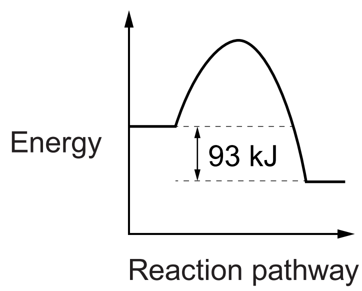


The bond energy of a H—H bond is 436 kJ. The energy needed to break **all** the bonds in the **reactants** is 2253 kJ.

(i) Calculate the energy needed to break a N≡N bond. [2]

Energy = ... kJ

(ii) The diagram below is the energy profile for the reaction.



Calculate the energy released when **all** the bonds in the ammonia molecules are formed. [1]

Energy = ... kJ

Examiner only

(b) The table below shows the percentage yield of ammonia under different pressure and temperature conditions.

Pressure (atm)	Temperature (°C)			
	200	300	400	500
	Percentage yield of ammonia			
100	81.7	52.5	25.2	10.6
200	89.0	66.7		18.3
400	94.6	79.7	55.4	31.9
1000	98.3	92.6	79.8	57.5

(i) Describe the relationship between the percentage yield of ammonia and the temperature. [1]

....

(ii) **Circle** the most likely percentage yield of ammonia if the process were carried out at 200 atm and 400 °C. [1]

20 % **30 %** **40 %** **50 %**

Examiner only

(c) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

Compound	Flame test	Add $NaOH(aq)$	Add $HCl(aq)$	Add $BaCl_2(aq)$	Add $AgNO_3(aq)$
A	no colour	pungent gas given off turns damp red litmus paper blue	no reaction	no reaction	white precipitate forms
B	green flame	blue precipitate forms	no reaction	white precipitate forms	no reaction
C	lilac flame	no reaction	gas given off turns limewater milky	no reaction	no reaction

Use the information provided to identify compounds **A**, **B** and **C**. [3]

A ....

B ....

C ....

8

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2					Group						3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3410UB0-1

S23-3410UB0-1

MONDAY, 22 MAY 2023 – MORNING

CHEMISTRY – Unit 2:

Chemical Bonding, Application of Chemical Reactions and Organic Chemistry

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**11**	
3.	**9**	
4.	**7**	
5.	**11**	
6.	**8**	
7.	**9**	
8.	**10**	
9.	**6**	
Total	**80**	

3410UB01
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6**(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

Examiner only

8. (a) According to the Journal of the British Dental Association, the increase in consumption of fruit juices and other acidic drinks is believed to be one of the leading causes of dental erosion in children and adolescents.

Many fruit juice drinks contain more than one type of acid. Citric acid can be used as a natural preservative and provides a sour taste. Ascorbic acid (vitamin C) is a water-soluble vitamin that must be consumed regularly to ensure proper body function. Citrus fruits, as well as tomatoes and other fresh vegetables, are good sources of vitamin C.

The table below shows information about the content of different fruit juice drinks.

Fruit juice drink	pH	Water (%)	Citric acid (%)	Ascorbic acid (mg / 100 g)	Sugar (%)
lime	2.2	76.4	4.80	29	1.68
lemon	2.4	81.3	4.60	53	1.80
grapefruit	3.0	90.3	1.35	38	2.34
orange	3.6	87.4	0.96	50	4.85
tangerine	3.8	85.7	0.74	31	7.89

(i) Tick (✓) the **two** conclusions that can be drawn from the information. [2]

as pH increases, citric acid content decreases and sugar content increases	
as acidity decreases, ascorbic acid content decreases and water content decreases	
tomatoes are a good source of vitamin C and citric acid	
citrus fruits contain ascorbic acid and a natural preservative	

(ii) **Use information from the table** to suggest why citric acid content has a much greater effect on the pH than ascorbic acid content. [1]

....

....

Examiner only

(b) The concentration of a solution of sodium hydroxide can be determined by titrating with sulfuric acid. The equation for the reaction is shown.

$$H_2SO_4 \quad + \quad 2NaOH \longrightarrow Na_2SO_4 \quad + \quad 2H_2O$$

(i) Give the **ionic** equation for the formation of water in any neutralisation reaction. Include state symbols. [2]

....

(ii) $21.0\,cm^3$ of sulfuric acid with a concentration of $0.350\,mol/dm^3$ neutralised $25.0\,cm^3$ of the sodium hydroxide solution.

I. Calculate the number of moles of sulfuric acid used in the reaction. [1]

Number of moles = ... mol

II. Calculate the concentration of the sodium hydroxide solution. [2]

Concentration = ... mol/dm^3

Examiner only

(iii) During a similar titration reaction, 0.36 g of water was produced. Calculate the number of **molecules** of water produced in this reaction.

Give your answer in **standard form**. [2]

$A_r(H) = 1$ $A_r(O) = 16$

Avogadro's constant = 6.0×10^{23}

Number of molecules = ...

10